Energy Consumption Metrics Definition

Household Energy Consumption Analytics Dashboard

Document: Metrics Definition

Project: Household Energy Consumption Analytics Dashboard

Dataset: UCI Individual Household Electric Power Consumption (household_energy_cleaned.csv)

1. Objective

This document defines the key energy consumption metrics that will drive the dashboard's visualizations and analysis, as required by the Product Requirements Document (functional requirements F1-F5) and Project Charter (Objective: “Define and implement meaningful energy-use metrics”). Each metric below is defined against the fields already produced in household_energy_cleaned.csv: the timestamp column and seven electrical measurements (global_active_power_kw, global_reactive_power_kw, voltage_v, global_intensity_a, sub_metering_1_wh, sub_metering_2_wh, sub_metering_3_wh), together with the derived time features (Year, Month, Day, Hour, DayOfWeek, Season) already generated in the analysis notebook.

2. Consumer Pain Points This Dashboard Addresses

Consumers today mostly experience their electricity usage as a single lagging number on a monthly bill, with no visibility into what actually drove it. This creates several recurring, well-documented pain points that the metrics in this document are each designed to answer:

- “Why is my bill high this month?” - no way to see total or average usage broken out by period to compare against a personal baseline.
- “When am I actually using the most power?” - no visibility into peak hours or time-of-day habits, so there's no way to shift usage away from costly or wasteful periods.
- “Is this month's bill normal for the season, or did something change?” - no easy way to separate expected seasonal swings (heating/cooling) from a genuine change in behavior.
- “Which appliances are actually driving my usage?” - a single total gives no diagnostic detail on where the energy is going.
- “Is something wrong with an appliance?” - consumers typically only discover a stuck water heater, faulty appliance, or phantom load weeks later via an abnormally high bill, with no earlier warning.

Section 3 maps each metric to the specific pain point it resolves, in addition to the PRD functional requirement it satisfies.

3. Metrics Overview

#	Metric	Primary Goal / FR Addressed
1	Total Energy Consumption (kWh)	F5 - key energy-use metric (total consumption)
2	Average Consumption (Mean Active Power)	F5 - key energy-use metric (average consumption)
3	Peak-Demand Period	F4 - highlight peak-demand periods
4	Time-of-Day Load Profile	F1 / F2 - visualize and filter by hour of day
5	Seasonal / Monthly Consumption Trend	F2 / F3 - filter and compare by month/season
6	Weekday vs. Weekend Consumption	F3 - compare across defined dimensions
7	Sub-Metering Category Share	F3 - compare consumption across categories
8	Unusual Consumption / Anomaly Flag	F4 - identify unusual spikes/drops

4. Detailed Metric Definitions

Metric 1: Total Energy Consumption (kWh)

Description: The total electrical energy consumed by the household over a selected period (day, week, month, season, or custom range), expressed in kilowatt-hours (kWh). This is the headline consumption figure for any selected view.

Consumer Pain Point Addressed: Answers “How much electricity did I actually use?” - giving consumers a concrete usage figure they can compare period-over-period, instead of only seeing a dollar amount on a bill weeks later.

Calculation Method:

1. Filter household_energy_cleaned.csv to the selected date/time range using the timestamp column.
2. Convert each one-minute power reading to an energy value: energy_kwh = global_active_power_kw × (1/60), since each row represents one minute of minute-averaged power.
3. Sum energy_kwh across all rows in the selected range to obtain Total Energy Consumption (kWh) for that period.

Justification / Relevance to Project Goals: Directly satisfies F5 (present total consumption) and NFR2 (usability) by giving users a single, intuitive headline number that anchors every other view in the dashboard.

Metric 2: Average Consumption (Mean Active Power)

Description: The average rate of power draw (kW) over a selected period and granularity (hourly, daily, or monthly), representing typical household load rather than cumulative energy.

Consumer Pain Point Addressed: Answers “Is my usage normal?” - without a personal baseline, consumers have no reference point to judge whether a given day or month is typical, high, or low.

Calculation Method:

1. Group rows by the desired time bucket (Hour, Day, or Month) using the derived timestamp columns.
2. Compute the mean of global_active_power_kw within each group.
3. Optionally compute a rolling/moving average across buckets to smooth short-term noise for trend visualizations (F1).

Justification / Relevance to Project Goals: Satisfies F5's average-consumption requirement and provides the baseline against which peaks (Metric 3) and anomalies (Metric 8) are measured.

Metric 3: Peak-Demand Period

Description: The time period(s) with the highest level of household demand - identifying when the household draws the most power, both on a per-day basis and across the dataset as a whole.

Consumer Pain Point Addressed: Answers “When am I using the most electricity?” - consumers on time-of-use rates, or simply trying to cut costs, have no way to shift usage away from expensive or wasteful hours if they can't see when their peak draw actually happens.

Calculation Method:

1. Aggregate global_active_power_kw to hourly means (group by calendar date + Hour).
2. For a chosen period (day, week, or month), identify the maximum hourly mean value and the hour it occurs in.
3. Across all days, compute the distribution of “peak hour of day” (e.g., modal peak hour) and/or a 95th-percentile threshold on hourly means to define a general peak-demand band for the dashboard to highlight.

Justification / Relevance to Project Goals: Directly satisfies F4 (highlight peak-demand periods), one of the two explicitly named dashboard capabilities in the PRD.

Metric 4: Time-of-Day Load Profile

Description: The average power consumption for each hour of the day (0-23), aggregated across all days in the dataset or a selected sub-period, producing a 24-point "typical day" curve.

Consumer Pain Point Addressed: Answers "What does my daily routine actually cost me?" - many consumers don't realize how much idle overnight draw or late-evening habits contribute to their bill until they see the shape of their own day.

Calculation Method:

1. Group all rows by the Hour column (0-23).
2. Compute the mean (and optionally the standard deviation or percentile band) of global_active_power_kw within each hour group.
3. Render as a line or bar chart with Hour on the x-axis for the dashboard's time-series view (F1).

Justification / Relevance to Project Goals: Supports F1 (visualize consumption over time) and F2/F3 (filter and compare by hour of day), and is the visual basis most users will interact with first.

Metric 5: Seasonal / Monthly Consumption Trend

Description: Average and total consumption aggregated by month and by season (Winter, Spring, Summer, Fall), showing how usage shifts across the year.

Consumer Pain Point Addressed: Answers "Is this a seasonal swing or did something change?" - a bill spike in July or January is often just weather-driven heating/cooling load, but without a seasonal view consumers can't tell that apart from an actual behavior or equipment change.

Calculation Method:

1. Use the derived Month and Season columns already created in the analysis notebook.
2. Group by Month (or Season) and compute both mean(global_active_power_kw) and summed energy in kWh (using the Metric 1 conversion) per group.
3. Compare the resulting pattern across the dataset's ~4 years to check whether seasonal behavior is consistent year over year.

Justification / Relevance to Project Goals: Satisfies F2/F3 (filter/compare by month and season) and reflects the analytical relevance already identified in the Dataset Research document.

Metric 6: Weekday vs. Weekend Consumption Comparison

Description: A comparison of average and total consumption on weekdays versus weekends, capturing behavioral differences driven by occupancy and routine.

Consumer Pain Point Addressed: Answers "Does being home more actually change my usage?" - consumers often suspect that work-from-home days or weekends at home are driving cost but have no way to quantify or confirm it.

Calculation Method:

1. Derive an is_weekend boolean from the DayOfWeek column (Saturday/Sunday = weekend).
2. Group rows by is_weekend, optionally cross-tabulated with Hour for a weekday-vs-weekend load-curve overlay.
3. Compute mean active power and total energy (kWh) for each group.

Justification / Relevance to Project Goals: Satisfies F3 (compare consumption across defined dimensions) and adds a practically meaningful comparison beyond hour/month/season alone.

Metric 7: Sub-Metering Category Share

Description: The proportion of total metered household energy attributable to each of the three sub-metering circuits - kitchen (sub_metering_1), laundry room (sub_metering_2), and water heater/air-conditioning (sub_metering_3) - versus remaining unmetered "other" usage.

Consumer Pain Point Addressed: Answers “Which appliances are actually driving my bill?” - this is consistently the single biggest pain point homeowners report: a total kWh number offers no diagnostic detail on where to cut back.

Calculation Method:

1. Convert global_active_power_kw to a comparable watt-hour basis per row: total_active_energy_wh = global_active_power_kw × 1000 / 60.
2. Sum sub_metering_1_wh, sub_metering_2_wh, sub_metering_3_wh, and total_active_energy_wh over the selected period.
3. Compute “other” usage as: other_wh = total_active_energy_wh_sum − (sub_metering_1_wh_sum + sub_metering_2_wh_sum + sub_metering_3_wh_sum).
4. Express each of the four categories (kitchen, laundry, water heater/AC, other) as a percentage of total_active_energy_wh_sum.

Justification / Relevance to Project Goals: Uses the dataset's unique sub-metering fields to move beyond aggregate demand into category-level insight, directly supporting the PRD's stated Context of “informing conservation efforts” and F3's comparison requirement.

Metric 8: Unusual Consumption / Anomaly Flag

Description: Identification of time periods where consumption deviates significantly from the expected pattern for that hour of day, flagged as unusually high or low relative to typical behavior.

Consumer Pain Point Addressed: Answers “Is something wrong?” - consumers typically only discover a stuck water heater, faulty appliance, or phantom load weeks later via an abnormally high bill; an early, visible flag turns a costly surprise into an actionable alert.

Calculation Method:

1. Compute the hourly-aggregated series of global_active_power_kw as in Metric 4.
2. For each hour-of-day baseline, compute the interquartile range (Q1, Q3) or mean/standard deviation across all days.
3. Flag hourly readings above Q3 + 1.5 × IQR, or below Q1 − 1.5 × IQR, as anomalies - consistent with the IQR screening approach already applied in the Preliminary Inspection (which flagged 94,907 raw readings, 4.63%, as statistically unusual).
4. Optionally compute a per-bucket z-score as a continuous “unusualness” score rather than a binary flag, to support a severity-shaded view in the dashboard.

Justification / Relevance to Project Goals: Directly satisfies F4 (identify unusual consumption spikes/drops) and keeps the anomaly methodology consistent with the screening already performed and documented in the Preliminary Inspection deliverable.

5. Acceptance Criteria Check

Requirement	Status
At least 5 key energy consumption metrics defined	Met - 8 metrics defined (Section 4).
Clear, concise description for each metric	Met - each metric includes a Description.
Step-by-step calculation method outlined for each metric	Met - each metric includes numbered Calculation Method steps.
Justification for why each metric is relevant to the project goals	Met - each metric includes a Justification tied to specific PRD functional requirements.